Vent

Reading philosophy in the canteen, building a *manifesto of meaning.* Then two colleagues sat down.

THE CANTEEN
Jung. Nietzsche. AI. 7 hrs/day on ChatGPT. Searching for meaning in the noise.

THE FRUSTRATION
Same two. Same problems. Every week. I knew the issues. I was done listening.

THE SPARK
Coder friend showed me his AI business. Firework went off. Drove home, one hour, thinking about nothing else.

THE BUILD
Ran upstairs. Cleared the desk. Opened Claude. Typed every idea in my head.

"Let's go mate. Let's build it."

Darren Doherty
Bioprocess Associate, WuXi Biologics · BSc Immunology, Trinity College Dublin

90-hour weeks since. Full product built. Still on the floor full-time.

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THE COST OF SILENCE

The floor knows.
The facility pays.

€1–5M

per batch failure

biologics drug substance

40–60

man-hours per CAPA

investigation & documentation

90

day average to close

deviation-to-closure cycle

> Deviation systems are traceable. Names are attached. Nobody wants to be the person who shut down the batch. **So the floor stays silent** , and the same deviation happens three shifts in a row before anyone connects the dots.

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VENT

30 seconds to submit. Zero fear.

Confidential. Plain language. Any device on the floor. The AI does everything that currently takes QA teams weeks.

1 Operator submits

OBSERVATION

"pH dropped 0.3 units below setpoint during feed addition on BR-03. Night shift. Recovered after 2 hours but no alarm triggered on SCADA."

Confidential | Upstream | Night

2 AI analyses in seconds

- Cross-references against facility SOPs — SOP-UP-003 §4.2.1
- Classifies risk & regulatory flags — HIGH
- Generates root cause hypothesis
- Writes full CAPA package — owners + timelines
- Routes to the right people — QA · MSAT · Eng

3 QA reviews

E-signatures. Audit trail. 21 CFR Part 11. The AI triages , humans decide.

4 Director sees live

Trends. Patterns. Open CAPAs. Real-time , not a weekly email.

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THE GAP

Veeva charges €240K/yr. They can't do any of this.

Confidential floor reporting	No QMS does this	✓
AI SOP analysis	No QMS does this	✓
Auto-generated CAPA packages	No QMS does this	✓
Built for operators, not QA	No QMS does this	✓

> We don't replace Veeva. We're the layer underneath they're missing , the one that captures what's actually happening on the floor. **At a fraction of the cost.**

WHAT VENT SAVES ONE SITE

€21–144K

per year

30–100 CAPAs/yr × 40–60 hrs each
30% reduction pays for the licence

Them
€60–480K
/site /year

Vent
€35–60K
/site /year

Pharma QMS market
$3B SAM · 13.3% CAGR

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WHAT'S BUILT

This isn't a pitch.
It's a deployed product.

AI

Multi-agent analysis pipeline
5-node LangGraph: classify → analyse → route → write CAPAs. Grounded in real SOPs via RAG. 9 specialised agents with versioned prompts.

QA

QA Control Centre + 4-phase workflow
IDE-style workspace. SOP cross-reference inline. Electronic signatures. 22 pre-configured facility contacts with expertise-based routing.

DOC

SOP search + Document Builder
RAG-powered SOP queries in plain language. Full document authoring with AI assist, version control, and media attachments.

VOICE

Charlie , voice AI for the floor
Hands-free SOP queries. VAPI integration. Speech-to-text submission. 3-language support (EN/ZH/ES) , because manufacturing floors are multilingual.

COMPLIANCE-READY FROM DAY ONE

- ✓ **21 CFR Part 11 / EU Annex 11** , e-signatures with re-auth on every approval
- ✓ **Immutable audit trail** , SHA-256 checksums, timestamps, IP addresses
- ✓ **ALCOA+ data integrity** , passes inspection without further development
- ✓ **Automated eval framework** , testing agent accuracy on every release
- ✓ **6 roles, row-level security** , operator to director, enforced at database layer

Built solo. Evenings and weekends.
While working full-time on the floor.
No funding. No co-founder. No excuses.

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THE ASK

I need to leave the floor *and build this full-time.*

First pilot

CDMO or MNC site in Ireland. I work inside WuXi. Pfizer, Lilly, MSD, Regeneron , all within 2 hours. These are warm doors.

Validate for GMP

IQ/OQ/PQ package , the formal documentation regulators require. The compliance architecture is built. The paperwork isn't.

Scale the team

Technical co-founder with enterprise SaaS experience. I built the product. I need someone who's scaled one.

9 of the world's top 10 pharma companies manufacture in Ireland.
The product is deployed. The market is here. I live this problem every single shift.

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